Tomoki Nishiyama

https://nishi24yama80.github.io/website/

Education

Graham School, the University of Chicago — Chicago, USA
Non-degree Program — 2025 - Present
- Real Analysis and Environmental Economics

Graduate School of Economics, The University of Tokyo — Tokyo, Japan
Master's Program in Economics — 2024–2025 (on leave)
- First-year PhD Microeconomics, Macroeconomics, and Econometrics
- Measure Theory, Measure-Theoretic Mathematical Statistics, Measure-Theoretic Stochastic Processes, Graph Theory, and Computational Mathematics

Department of Economics, The University of Tokyo — Tokyo, Japan
B.S. in Economics — 2020 - 2024

Awards and Honors

- **Brandt Research Fellow Award**, An award for an EPIC predoc 2026.09 *Energy Policy Institute at UChicago*, *Award: $2,500*
- **Specially Selected Paper**, Thesis at the University of Tokyo 2024.03
- **First Prize**, Empirical Micro Conference at Keio University 2023.07

Working Papers

Inefficiencies of Revenue Sharing Auctions
2026.06 - Present

Production Function Estimation with a Fixed Point Approach
Extension of Trunschke and Judd (2024) — 2026.04 - Present

Work in Progress

Staying for Opportunity: Evidence from an In-Situ Housing Lottery in a Low-Income Community
with Andrew S. Griffen, Yukichi Mano, Yasuyuki Sawada, and Ze Wang

Screening in Public Procurement
with Ze Wang

Patent

Machine Learning Model for Predicting Vacant Houses Using Administrative Data and Water Usage
Japan Patent ID: JP-2024-018823 — 2024.02

Machine Learning Model for Predicting Vacant Houses Using Administrative Data and Utility Consumption Data (Water and Gas)
Japan Patent ID: JP-7137267 — 2022.09

Job Experience

Pre-doctoral Research Professional for Koichiro Ito | Chicago, USA 2025.9 - Present
- Brandt Research Fellow (2026.9 - Now)
- Research Professional at Energy Policy Institute, the University of Chicago
- Becker Friedman Institute

Research Assistant for Yasuyuki Sawada | Tokyo, Japan 2022.07 - 2025.8
- Engaged in six field surveys in Manila, the Philippines.
- Prepared survey forms and manage the surveys with local collaborators in Manila.
- Constructed dataset from the field-survey results.
- Analyzed data with causal inference and deep-learning methods.

Research Assistant at Hirofumi Uzawa International Institute | Tokyo and Kyoto, Japan 2023.07 - 2025.8
- Constructed a model of environment economics and macroeconomics.
- Wrote academic articles for the general public.
- Made academic presentations at Kyoto University.

Research Assistant for Taisuke Nakata | Tokyo, Japan 2021.07 - 2021.12
- Constructed epidemic-macro models.
- Numerical simulations with Matlab.
- Prepared policy reports for Ministry of Health, Labour and Welfare.

GIS Data Analyst at Microbase Inc. | Tokyo Japan 2020.09 - 2023.07
- Data cleaning of administrative data and utility consumption data
- Geo-spatial analysis of the administrative data
- Development of machine-learning models of predicting vacant houses
- Optimization of future energy infrastructure distribution
- Prepare reports and presentations summarizing research findings
- Develop web-application

Presentations

- Special Symposium "Evolution of Social Common Capital" (Kyoto University) 2024.03
- Japanese Association for Development Economics, Young Conference (Kyoto University) 2023.11
- Empirical Micro Conference (Keio University) 2023.07
- Korea-Japan Inter University Conference (Online) 2022.11

Skills

Languages: Japanese (Native), English (Fluent), Mandarin Chinese (Intermediate).

Programming: Python, R, Julia, Matlab, LaTeX, HTML, SQL, GIS (GeoPandas, GDAL, QGIS, PostGIS), Git.